

Kevin Chia-Ming Chang

PH.D. CANDIDATE | FORMAL METHODS, VERIFICATION, MACHINE LEARNING

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SUMMARY

Ph.D. candidate at UC Berkeley building scalable formal-methods and machine-learning research for safety-critical systems. Experience spans neural network verification, contract-based design, optimization, and interpretable controller synthesis, with a focus on translating theory into solver-backed algorithms and rigorous empirical evaluation.

SKILLS

Research Areas	Formal verification, interpretable ML, contract-based design, reinforcement learning
Methods	SMT, MILP, assume-guarantee reasoning, symbolic abstraction, safe reachability
Programming	Python, SystemVerilog, MATLAB, C++
ML / Data	PyTorch, scikit-learn, reinforcement learning, semantic segmentation, domain adaptation
Tools	Gurobi, CVX, JasperGold, Cadence Xcelium, Genus, VCS

RESEARCH EXPERIENCE

DesCyPhy Lab, Prof. Pierluigi Nuzzo

Graduate Student Researcher

Berkeley, CA

01/2021 – Present

Neural Controller Transformation for Formal Verification

- Developed a **provably exact transformation** from ReLU neural-network controllers to soft decision-tree representations, making solver-based verification more scalable and interpretable.
- Proved correctness and demonstrated closed-loop safety certification on CartPole, MountainCar, and CarRacing benchmarks; resulted in a **CDC 2023 publication**.

Robustness Contracts for Learning-Enabled CPS

- Introduced **contract-based robustness specifications** that decouple plant dynamics from neural controllers for compositional safety verification.
- Designed an **SMT-based refinement algorithm** that incrementally tightens layer-wise abstraction bounds to accelerate safe-reachability analysis under bounded uncertainty.

Assume-Guarantee Contract Optimization

- Formulated **MILP-based contract-refinement** techniques to automatically synthesize tighter assume-guarantee specifications.
- Improved realizability and reduced verification complexity in multi-agent autonomous-driving scenarios.

Deep RL Controller Distillation for Verifiability

- Developed a controller-distillation framework that extracts compact, interpretable control laws from deep RL policies.
- Applied optimization-based refinement to preserve task performance while improving closed-loop stability and formal verifiability; resulted in an **Allerton 2022 publication**.

Electronic Design Automation Lab, Prof. Yao-Wen Chang

Undergraduate Researcher

Taipei, Taiwan

09/2018 – 02/2020

Routing and Optimization for EDA Contests

- Implemented an initial detailed-routing engine with concurrent track assignment and multithreaded routing; handled designs with up to **1M nets** using less than **64 GB** of memory in 10 hours on most benchmarks.
- Built a routability-driven hybrid global router and a system-level FPGA router with TDM optimization; contributed to **2nd place at ACM ISPD 2019, Top 5 / Honorable Mention at ICCAD 2019**, and an **ICCAD 2021 publication**.

EDUCATION

University of California, Berkeley

Ph.D. in Electrical and Computer Engineering (GPA: 3.91/4.00)

Berkeley, CA

01/2021 – Present

- Research focus: formal verification, ML-enabled cyber-physical systems, optimization, and contract-based design

Lab: DesCyPhy Lab, The Donald O. Pederson Center for Electronic Systems Design

National Taiwan University

B.S. in Electrical Engineering (GPA: 3.67/4.00, Last 60: 4.00/4.00)

Taipei, Taiwan

09/2015 – 01/2020

- Relevant coursework: physical design, integrated circuit design, computer architecture, machine learning, deep learning for computer vision

- Undergraduate research in EDA and formal methods

ADDITIONAL EXPERIENCE

Apple

Formal Verification Engineer Intern

Cupertino, CA

05/2026 – 08/2026

- Developed and debugged **formal properties** for CPU microarchitecture blocks, targeting control-flow correctness.
- Analyzed proof failures by refining assertions, assumptions, and coverage targets to distinguish design issues from environmental constraints.

University of California, Berkeley

Berkeley, CA

Graduate Student Instructor of Introduction to Design Automation

01/2026 – 05/2026

- Developed **end-to-end instructional examples** demonstrating the **EDA toolchain workflow** across synthesis, simulation, and verification.
- Supported course delivery for design automation topics spanning RTL design, logic synthesis, timing analysis, and verification workflows.

University of Southern California

Los Angeles, CA

Graduate Student Instructor (3×) of Computer Architecture

01/2022 – 05/2023

- Developed **instructional examples** illustrating processor pipelines and the transition from in-order to out-of-order execution.
- Supported students in reasoning about performance tradeoffs across pipelining, hazards, caching, and memory hierarchy

Institute of Information Science, Academia Sinica

Taipei, Taiwan

Software Researcher

06/2018 – 09/2018

- Implemented a semi-supervised domain-adaptation pipeline for semantic segmentation using unpaired image translation and segmentation modeling.
- Improved mean IoU on the Cityscapes benchmark by combining segmentation training with CycleGAN-generated domain information.

SELECTED PROJECT

Sign Language Translation System

[C++/Python/PyTorch/Scikit-Learn]

Arm Design Contest

09/2018

- Built a real-time sign-language translation system using a single 9-DOF IMU for gesture recognition.
- Won **2nd Place** and the **Best Popularity Award** in the Arm Design Contest.

SELECTED PUBLICATIONS

1. Kevin Chang, Nathan Dahlin, Rahul Jain, and Pierluigi Nuzzo, “Equivalent and Compact Representations of Neural Network Controllers With Decision Trees,” *IEEE Transactions on Automatic Control*, pp. 1–14, 2026.
2. Kevin Chang, Nathan Dahlin, Rahul Jain, and Pierluigi Nuzzo, “Exact and Cost-Effective Automated Transformation of Neural Network Controllers to Decision Tree Controllers,” in *Proc. IEEE Conference on Decision and Control (CDC)*, pp. 7843–7848, 2023.
3. Nathan Dahlin, Kevin Chang, K. C. Kalagarla, Rahul Jain, and Pierluigi Nuzzo, “Practical Control Design for the Deep Learning Age: Distillation of Deep RL-Based Controllers,” in *Proc. Allerton Conference on Communication, Control, and Computing*, 2022.
4. Jair Certo, Kevin Chang, Nuno C. Martins, Pierluigi Nuzzo, and Yasser Shoukry, “Passivity Tools for Hybrid Learning Rules in Large Populations,” *Automatica*, 2026.
5. Wei-Kai Liu, Ming-Hung Chen, Chia-Ming Chang, Chen-Chia Chang, and Yao-Wen Chang, “Time-Division Multiplexing Based System-Level FPGA Routing,” in *Proc. IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2021.

SELECTED AWARDS

Outstanding Performance Scholarship, National Taiwan University

Taipei, Taiwan

Awarded for outstanding academic performance in electrical engineering

11/2019

- Recognized for strong academic performance and research contributions during undergraduate study in electrical engineering.

Top 5 and Honorable Mentions, Cad Contest Problem C

Westminster, USA

LEF/DEF-based open-source global router

11/2019

- Recognized in an ACM/IEEE contest field of 100+ teams.

2nd Place, ACM ISPD Contest

San Francisco, USA

Initial detailed routing

04/2019

- Placed 2nd among 33 international teams in a leading EDA routing competition.

2nd Place and Best Popularity Award, Arm Design Contest

Taipei, Taiwan

Sign language translation system

11/2018

- Placed in the top 3 among 130+ teams in a hardware and embedded-systems design competition.